OTHERS					
TC ID	**Title**	**Pre-condition**	**Steps to Reproduce**	**Expected Result**	**Actual Result**
TC-01	App to background during gameplay	Game running	Press Home	App pauses safely	
TC-02	Resume app from background	App in background	Reopen app	Game state restored	
TC-03	Background during challenge creation	Create challenge open	Minimize app	No data loss	
TC-04	Background during QR scan	QR scan active	Press Home	App resumes safely	
TC-05	Background during Results view	Results open	Switch app	State preserved	
TC-06	Incoming call interruption	Game running	Receive call	App resumes correctly	
TC-07	Notification interruption	App active	Notification received	No UI break	
TC-08	Lock screen during gameplay	Game running	Lock phone	Resume correctly	
TC-09	App killed by OS	App background	OS kills app	Graceful relaunch	
TC-10	Background > 30 mins	App background	Wait 30 mins	No crash on return	
TC-11	Background audio behavior	Music ON	Minimize app	Audio pauses/resumes as expected	
TC-12	Background + Sound OFF	Sound OFF	Minimize	No sound leakage	
TC-13	Background + Timer running	Timer enabled	Minimize	Timer pauses/resumes correctly	
TC-14	Background + network loss	App background	Disable network	Stable return	
TC-15	Background + rotation	App background	Rotate device	UI intact	
TC-16	App switch stress	Switch apps rapidly	Repeat 10x	No crash	
TC-17	Split screen mode	Split screen	Use app	Layout stable	
TC-18	Picture-in-picture conflict	Video app active	Open game	No conflict	
TC-19	Background + low battery	Battery <15%	Resume app	App stable	

TC-20	Background restore accuracy	Game mid-state	Resume	Exact state restored	
TC-21	Fresh install storage	New install	Check app size	Within expected limit	
TC-22	Storage after gameplay	Play 10 games	Check storage	Reasonable growth	
TC-23	Storage after 25 results	Max results	Check storage	No abnormal spike	
TC-24	Storage after delete results	Results deleted	Check storage	Storage reclaimed	
TC-25	Storage after challenges created	Create 25 challenges	Check storage	Stable usage	
TC-26	Storage after QR scans	Scan multiple QRs	Check storage	No junk files	
TC-27	Cache growth	Long usage	Check cache	Cache controlled	
TC-28	Clear cache manually	Clear cache	Reopen app	App works	
TC-29	Clear app data	Clear data	Open app	Fresh launch	
TC-30	App uninstall cleanup	Uninstall app	Check storage	No leftover files	
TC-31	Reinstall behavior	Reinstall app	Open	Fresh state	
TC-32	Clipboard storage usage	Paste codes	Monitor	No persistent data	
TC-33	Screenshot heavy usage	Take many screenshots	Monitor storage	App unaffected	
TC-34	Low storage device	<500MB free	Launch app	App works	
TC-35	Storage permission denial	Deny storage	Upload QR	Graceful error	
TC-36	Storage permission grant	Grant later	Retry upload	Works	
TC-37	Large gallery selection	Select large image	Upload QR	No crash	
TC-38	Corrupt image file	Upload corrupt QR	Upload	Error shown	
TC-39	Background storage cleanup	App idle	Wait	No leaks	
TC-40	Storage after long idle	Idle 24 hrs	Check	Stable	
TC-41	Continuous gameplay	App open	Play 30 mins	No lag	
TC-42	Rapid letter rotations	Game active	Rotate fast	Smooth UI	
TC-43	Rapid submit taps	Game active	Tap submit fast	No crash	
TC-44	Rapid challenge creation	Create challenges	Repeat 20x	Stable	

TC-45	Rapid delete actions	Delete items	Repeat	No freeze	
TC-46	Rapid navigation	Switch screens	50 times	Stable	
TC-47	Multi-hour usage	App open	2+ hrs	No memory leak	
TC-48	Stress sound toggle	Toggle sound	50 times	No audio bug	
TC-49	Stress music toggle	Toggle music	50 times	No overlap	
TC-50	Stress settings changes	Change settings	Repeated	Saved correctly	
TC-51	Rapid QR paste	Paste codes	20 times	Stable	
TC-52	Rapid invalid input	Paste junk	Repeated	Proper errors	
TC-53	Heavy scrolling	Long lists	Scroll fast	Smooth	
TC-54	Stress orientation change	Rotate device	Rapid	No UI break	
TC-55	Stress back navigation	Back pressed	Rapid	No exit crash	
TC-56	Low-end device stress	Low RAM device	Heavy use	Acceptable	
TC-57	High-end device stress	Flagship device	Heavy use	Smooth	
TC-58	Background + foreground stress	App switching	Rapid	Stable	
TC-59	Battery saver ON	Saver enabled	Play game	App usable	
TC-60	Thermal throttling	Device hot	Play	App responsive	
TC-61	25 result entries	Max reached	Add more	Handled	
TC-62	25 challenge entries	Max reached	Add more	Handled	
TC-63	Challenge history stress	Scroll/delete	Repeated	Stable	
TC-64	Results expand stress	Expand/collapse	Repeated	No lag	
TC-65	Timer ON stress	Timer enabled	Long play	Accurate	
TC-66	Timer OFF stress	Timer disabled	Play	No UI gap	
TC-67	Multiple notifications	Notifications spam	Use app	Stable	
TC-68	Memory pressure	Other apps heavy	Use app	No crash	
TC-69	OS background cleanup	OS aggressive	Resume app	Graceful	
TC-70	App idle overnight	App idle	Open next day	Works	
TC-71	App cold start time	App closed	Launch	Fast launch	
TC-72	Warm start time	App background	Open	Instant	

TC-73	FPS stability	Gameplay	Observe	Smooth	
TC-74	UI jank detection	Animations	Observe	No stutter	
TC-75	Network OFF launch	Network OFF	Open app	Works	
TC-76	Network toggle mid-game	Game running	Toggle network	No crash	
TC-77	Offline QR paste	Network OFF	Paste code	Proper handling	
TC-78	Airplane mode	Airplane ON	Use app	Stable	
TC-79	Permission revoke mid-use	Revoke storage	Upload QR	Error shown	
TC-80	Permission restore	Restore permission	Retry	Works	
TC-81	App update retention	Update app	Open	Data intact	
TC-82	Version mismatch data	Old data	New version	Compatible	
TC-83	Language change	System language change	Open app	UI intact	
TC-84	Font scaling extreme	200% font	Open	No overlap	
TC-85	Dark mode compatibility	System dark	Open app	Proper theme	
TC-86	Light mode compatibility	Light mode	Open app	Proper theme	
TC-87	Screen notch handling	Notch device	Observe	No cut	
TC-88	Gesture navigation	Gesture mode	Navigate	Works	
TC-89	Three-button nav	Buttons enabled	Navigate	Works	
TC-90	App crash recovery	Force crash	Relaunch	Clean start	
TC-91	ANR prevention	Long tasks	Observe	No ANR	
TC-92	Battery drain check	30 mins play	Monitor	Acceptable drain	
TC-93	CPU usage check	Gameplay	Monitor	Reasonable	
TC-94	RAM usage check	Heavy use	Monitor	No leak	
TC-95	Accessibility TalkBack	TalkBack ON	Navigate	Readable	
TC-96	Screen reader focus	TalkBack	Focus UI	Logical	
TC-97	Long challenge names	Max length	Save	Handled	
TC-98	Special characters input	Symbols	Input	Sanitized	
TC-99	App force stop	Force stop	Relaunch	Stable	
TC-100	Overall stability pass	Full usage	Observe	No critical issues	